

Lab 10

Submission by: **May 9th - 4pm**

1) [0.2 points]

Let X be the set $\{1, 2, 3, 4, 5\}$ and Y be the set $\{6, 7, 8, 9, 10\}$. We describe the functions $f: X \rightarrow Y$ and $g: X \rightarrow Y$ in the following tables. Answer each part and give a reason for each negative answer.

n	$f(n)$	n	$g(n)$
1	6	1	10
2	7	2	9
3	6	3	8
4	7	4	7
5	6	5	6

- Is f injective?
- Is f surjective?
- Is f bijective?
- Is g injective?
- Is g surjective?
- Is g bijective?

2) [0.3 points]

Let T be the set of all infinite sequences over $\{0,1,2\}$. Show that T is uncountable using a proof by diagonalisation.

3) [0.3 points]

Let $O = \{(i, j, k) \mid i, j, k \in \mathbb{N}\}$. Show that O is countable.

4) [0.3 points]

Prove that it is decidable to determine whether a Turing machine ever writes a non-blank symbol when started on a blank tape.

5) [0.3 points]

Is the question “Does a Turing machine have at least 481 states” decidable? Explain your answer.

6) [0.6 points]

Let $A = \{ \langle M \rangle \mid M \text{ is a DFA that doesn't accept any string containing an even number of 1s} \}$. Show that A is decidable.